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ELECTRONIC DEVICE CASING, RACK ASSEMBLY AND IMMERSION COOLING SYSTEM

Abstract

An electronic device casing includes a first plate and two second plates. The first plate includes at least one embossing structure. Times of an average moment of inertia of cross sections of the first plate in a first direction are not smaller than 1.14. The two second plates are connected to the first plate and form an accommodation space with the first plate.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This non-provisional application claims priority under 35 U.S.C. § 119 (a) on Patent Application No(s). 113106488 filed in Taiwan, R.O.C. on Feb. 23, 2024, the entire contents of which are hereby incorporated by reference.

TECHNICAL FIELD

[0002] The disclosure relates an electronic device casing, a rack assembly and an immersion cooling system.

BACKGROUND

[0003] In general, a bottom plate of a casing of a server is mainly used to support various electronic components in the server. Therefore, in order to enable the bottom plate to have a sufficient structural strength for supporting those electronic component while not being deformed, the bottom plate of the casing of the server is required to have a sufficient thickness, but this may against lightweight requirement and carbon reduction requirement. As a result, how to achieve the lightweight requirement and the carbon reduction requirement while satisfying the structural strength of the bottom plate of the casing is one of the crucial topics in this field.

SUMMARY

[0004] One embodiment of the disclosure provides an electronic device casing. The electronic device casing includes a first plate and two second plates. The first plate includes at least one embossing structure. Times of an average moment of inertia of cross sections of the first plate in a first direction are not smaller than 1.14. The two second plates are connected to the first plate and form an accommodation space with the first plate.

[0005] Another embodiment of the disclosure provides a rack assembly. The rack assembly includes a rack and an electronic device casing. The electronic device casing is horizontally or vertically mounted in the rack and includes a first plate and two second plates. The first plate includes at least one embossing structure. Times of an average moment of inertia of cross sections of the first plate in a first direction are not smaller than 1.14. The two second plates are connected to the first plate and form an accommodation space with the first plate.

[0006] Still another embodiment of the disclosure provides an immersion cooling system. The immersion cooling system includes a container and an electronic device. The container is configured to accommodate a coolant. The electronic device is configured to be disposed in the container and at least partially immersed in the coolant. The electronic device includes an electronic device casing and a motherboard. The electronic device casing is configured to be mounted in the container and includes a first plate. The first plate includes at least one embossing structure, and times of an average moment of inertia of cross sections of the first plate in a first direction are not smaller than 1.14. The motherboard is fixed on the first plate and is provided with a heat generating module.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] The present disclosure will become better understood from the detailed description given herein below and the accompanying drawings which are given by way of illustration only and thus are not intending to limit the present disclosure and wherein:

[0008] FIG. 1 is a perspective view of a rack assembly according to a first embodiment of the disclosure;

[0009] FIG. 2 is a perspective view of an electronic device casing in FIG. 1;

[0010] FIG. 3 is a top view of the electronic device casing in FIG. 2;

[0011] FIG. 4 is a curve chart showing a relationship between times of average moment of inertia

and improvement rate of downward bending deformation;

[0012] FIG. 5 is a partially enlarged view of a unit of a first plate along a first direction in FIG. 3;

[0013] FIG. 6 is a partially enlarged view of a unit of the first plate along a second direction in FIG. 3;

[0014] FIG. 7 shows a cross section in u-v coordinate system;

[0015] FIG. 8 shows the cross section in x-y coordinate system;

[0016] FIG. 9 is a top view of an electronic device casing according to a second embodiment of the disclosure;

[0017] FIG. 10 is a partially enlarged view of a unit of a first plate along a first direction in FIG. 9;

[0018] FIG. 11 is a partially enlarged view of a unit of the first plate along a second direction in FIG. 9; and

[0019] FIG. 12 is a cross-sectional view of an immersion cooling system according to a third embodiment of the disclosure.

DETAILED DESCRIPTION

[0020] In the following detailed description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the disclosed embodiments. It will be apparent, however, that one or more embodiments may be practiced without these specific details. In other instances, well-known structures and devices are schematically shown in order to simplify the drawing.

[0021] In addition, the terms used in the present disclosure, such as technical and scientific terms, have its own meanings and can be comprehended by those skilled in the art, unless the terms are additionally defined in the present disclosure. That is, the terms used in the following paragraphs should be read on the meaning commonly used in the related fields and will not be overly explained, unless the terms have a specific meaning in the present disclosure.

[0022] Referring to FIGS. 1 to 3, FIG. 1 is a perspective view of a rack assembly according to a first embodiment of the disclosure, FIG. 2 is a perspective view of an electronic device casing in FIG. 1, and FIG. 3 is a top view of the electronic device casing in FIG. 2.

[0023] In this embodiment, the rack assembly 1 includes a rack 10 and an electronic device casing 20. The rack 10 is, for example, a server rack. The electronic device casing 20 is, for example, a server casing, and the electronic device casing 20 is mounted in the rack 10.

[0024] The electronic device casing 20 includes a first plate 21 and two second plates 22. The first plate 21 includes a support portion 211 and a plurality of embossing structures 212. The support portion 211 has a support surface 2111. The second plates 22 respectively stand on two opposite sides of the support surface 2111 of the support portion 211 of the first plate 21. The first plate 21 and the second plates 22 together form an accommodation space S, and the accommodation space S is, for example, configured to accommodate electronic components of a server (not shown), such as a motherboard, hard disk drives and fans. The embossing structures 212 are formed on the support surface 2111 and are arranged in an array. The embossing structures 212 may be formed on the support surface 2111 by performing an embossing process along a direction from one surface of the support portion 211 facing away from the support surface 2111 towards the support surface 2111. For example, each of the embossing structures 212 includes a main portion 2121 and two branch portions 2122. The main portion 2121, for example, has four recesses 21211, two of the recesses 21211 are located at one side of the main portion 2121 and are spaced apart from each other, and the other two of the recesses 21211 are located at another side of the main portion 2121 and are spaced apart from each other. As shown in FIG. 3, the recesses 21211 are symmetrically recessed inwards from two opposite long sides of the main portion 2121 which substantially has a rectangular shape. Each of the two branch portions 2122 includes two end parts 21221 and a connection part 21222 connected to the two end parts 21221. The two end parts 21221 of one of the branch portions 2122 respectively correspond to two of the recesses 21211 located at one side of the main portion 2121, and the two end parts 21221 of the other one of the branch portions 2122

respectively correspond to the other two of the recesses **21211** located at another side of the main portion **2121**. As shown in FIG. 3, the connection parts **21222** are spaced apart from the main portion **2121**, and the end parts **21221** respectively extend into the recesses **21211** of the main portion **2121**. In other words, the branch portions **2122** may be symmetrically formed at two opposite sides of the main portion **2121** which substantially has a rectangular shape.

[0025] In this embodiment, the electronic device casing **20** is horizontally placed into the rack **10**, and the second plates **22** of the electronic device casing **20** located opposite to each other are fixed to two opposite sides of the rack **10**. When there are electronic components uniformly placed on the first plate **21**, the downward bending deformation may mainly occur at the central portion of the first plate **21**. Cross sections of the first plate **21** along a first direction **D1** and a second direction **D2** can reflect the deformation of the first plate **21** more obviously, where the first direction **D1** is, for example, parallel to the second plate **22**, and the second direction **D2**, for example, intersects the first direction **D1**. For example, the second direction **D2** is non-parallel and non-perpendicular to the first direction **D1**. In other embodiments, the second direction **D2** may be at an angle to a diagonal line of an embossing unit of the first plate **21**, and the angle may be ± 2 degrees. In a case that the embossing unit of the first plate **21** is in a square shape, an angle θ between the second direction **D2** and the first direction **D1** may fall within a range from 43 degrees to 47 degrees, such as 45 degrees. As shown in FIG. 3, in one of the embossing structures **212**, the connection parts **21222** and the main portion **2121** may extend along a direction perpendicular to the first direction **D1**, and the end parts **21221** respectively extend towards the recesses **21211**.

[0026] By simulating a force to be applied on a plate with embossing structures and a flat plate without any embossing structure, a relationship between times of average moment of inertia and improvement rate of downward bending deformation of the plate with the embossing structures relative to the flat plate without any embossing structure may be summarized in, for example, FIG. 4, where FIG. 4 is a curve chart showing the aforementioned relationship. Times of the average moment of inertia of the plate with the embossing structures relative to the flat plate without any embossing structure represent that the ratio of the average moment of inertia of cross sections of the plate with the embossing structures in one direction to the average moment of inertia of cross sections of the flat plate without any embossing structure in such direction. As a result, when the times of the average moment of inertia of the cross sections of the plate with the embossing structures in such direction fall within the range from 1.14 to 1.18 (i.e., not smaller than 1.14 and not greater than 1.18), the improvement rate of downward bending deformation may be up to 4% to 10%. In some embodiments, the times of the average moment of inertia of the cross sections of the plate with the embossing structures in such direction is not smaller than 1.14. In some embodiments, the times of the average moment of inertia of the cross sections of the plate with the embossing structures in such direction is not greater than 1.18. In some embodiments, the times of the average moment of inertia of the cross sections of the plate with the embossing structures in such direction is not smaller than 1.14 and not greater than 1.18. When the times of the average moment of inertia of the cross sections of the plate with the embossing structures in such direction fall within the range from 1.15 to 1.17 (i.e., not smaller than 1.15 and not greater than 1.17), the improvement rate of downward bending deformation may be up to 8% to 10%. In some embodiments, the times of the average moment of inertia of the cross sections of the plate with the embossing structures in such direction is not smaller than 1.15. In some embodiments, the times of the average moment of inertia of the cross sections of the plate with the embossing structures in such direction is not greater than 1.17. In some embodiments, the times of the average moment of inertia of the cross sections of the plate with the embossing structures in such direction is not smaller than 1.15 and not greater than 1.17.

[0027] In this embodiment, the times of the average moment of inertia of the cross sections of the first plate **21** in the first direction **D1** and the second direction **D2** fall within the range from 1.14 to 1.18 (i.e., not smaller than 1.14 and not greater than 1.18). The times of the average moment of

inertia of the cross sections of the first plate **21** in the first direction **D1** represent that, in the first direction **D1**, the ratio of the average moment of inertia of the cross sections of the first plate **21** to the average moment of inertia of the cross sections of a flat plate having the same size as the first plate **21** but without any embossing structure. Similarly, the times of the average moment of inertia of the cross sections of the first plate **21** in the second direction **D2** represent that, in the second direction **D2**, the ratio of the average moment of inertia of the cross sections of the first plate **21** to the average moment of inertia of the cross sections of a flat plate having the same size as the first plate **21** but without any embossing structure.

[0028] In this embodiment, the embossing structures **212** are formed on the support surface **2111** of the support portion **211** of the first plate **21**, and the times of the average moments of inertia of the cross sections of the first plate **21** in the first direction **D1** and the second direction **D2** fall within the range from 1.14 to 1.18 (i.e., not smaller than 1.14 and not greater than 1.18), which can achieve lightweight requirement and carbon reduction requirement while satisfying the structural strength of the first plate **21**. Preferably, the times of the average moments of inertia of the cross sections of the first plate **21** in the first direction **D1** and the second direction **D2** may fall within the range from 1.15 to 1.17 (i.e., not smaller than 1.15 and not greater than 1.17), which can further increase the structural strength of the first plate **21** so as to reduce the downward bending deformation amount of the first plate **21**.

[0029] The following paragraphs will take some examples for illustration. Referring to FIGS. 3, 5 and 6, FIG. 5 is a partially enlarged view of a unit of a first plate along a first direction in FIG. 3, and FIG. 6 is a partially enlarged view of a unit of the first plate along a second direction in FIG. 3.

[0030] In order to clearly illustrate the average moment of inertia of the cross sections of the first plate **21** in the first direction **D1** and the second direction **D2**, the following descriptions take a unit **U1** (e.g., shown in FIG. 5) of the first plate **21** in the first direction **D1** and a unit **U2** (e.g., shown in FIG. 6) of the first plate **21** in the second direction **D2** for instance.

[0031] In the first direction **D1**, one unit **U1** of the first plate **21** is shown in FIG. 5. Assuming that the unit **U1** shown in FIG. 5 is a square portion of 50 mm×50 mm, the thickness of the support portion **211** is 1 mm, the heights of the main portions **2121** and the branch portions **2122** of the embossing structures **212** formed on the support surface **2111** are 0.2 mm, and the unit **U1** is cut into ten cross sections 5-1~5-10 along the first direction **D1** with the same intervals, the moment of inertia of each of the cross sections 5-1~5-10 can be calculated by a computer software based on formula of moment of inertia, $I = \int y^2 dA$, and is listed in Table 1.

[0032] In order to explain how to calculate the moment of inertia of one cross section, the following explanations are given as the cross section is simplified to be in a triangular shape. Referring to FIGS. 7 and 8, FIG. 7 shows a cross section in u-v coordinate system, and FIG. 8 shows the cross section in x-y coordinate system. The first step is to calculate an area of the cross section shown in FIG. 7, which can be obtained from the following equations:

$$[00001] A = \int dA = \int \int dv du = \int_0^b v du = \int_0^b (h - \frac{h}{b}u) du = hu - \frac{h}{2b}u^2 \text{ .Math. } \int_0^b = \frac{1}{2}bh \text{ .}$$

Then, the second step is to calculate the centroid of the cross section in v-axis direction, which can be obtained from the following equation,

$$[00002] v_c = \frac{\int v dA}{A},$$

where

$$[00003] \int v dA = \int \int v dv du = \int_0^b uv dv = \int_0^h (bv - \frac{b}{h}v^2) dv = \frac{bv^2}{2} - \frac{b}{3h}v^3 \text{ .Math. } \int_0^h = \frac{1}{6}bh^2,$$

and thus

$$[00004] v_c = \frac{\int v dA}{A} = \frac{\frac{1}{6}bh^2}{\frac{1}{2}bh} = \frac{h}{3}.$$

Then, as shown in FIG. 8, the third step is to upshift the coordinate system to make x-axis of the upshifted coordinate system pass through the centroid of the cross section. Then, the fourth step is to calculate the moment of inertia of the cross section according to the shifted coordinate system,

which can be obtain from the following equations:

[00005]

$$I_{xx} = \int y^2 dA = \int xy^2 dy = \int (\frac{2}{3}b - \frac{b}{h}y)y^2 dy = \int (\frac{2}{3}by^2 - \frac{b}{h}y^3)dy = \frac{2}{9}by^3 - \frac{b}{4h}y^4 \text{ .Math. } \frac{\frac{2}{3}b}{-\frac{1}{3}h} = \frac{1}{36}bh^3 ,$$

where “dA” is a small area in the cross section, and “y” is a distance from such small area to x-axis.

TABLE-US-00001 TABLE 1 Cross sections Moments of inertia (mm.sup.4) 5-1 4.698769 5-2

4.811584 5-3 4.930271 5-4 4.930271 5-5 4.811584 5-6 4.698769 5-7 4.811584 5-8 4.930271 5-9

4.930271 5-10 4.811584 Average 4.836496

[0033] As for a flat plate having the same size as the unit U1 but without any embossing structure, the average moment of inertia of the cross sections of the flat plate in the first direction D1 is about 4.166667 mm.sup.4. As a result, in the first direction D1, the times of the average moment of inertia of the cross sections 5-1-5-10 of the unit U1 relative to the average moment of inertia of the cross sections of the flat plate having the same size as the unit U1 but without any embossing structure is about 1.161. In other words, in the case that the average moment of inertia of the cross sections of the flat plate in the first direction D1 is 4.166667 mm.sup.4, the range from 1.14 to 1.18 times 4.166667 mm.sup.4 equals the range from 4.75000038 mm.sup.4 to 4.91666706 mm.sup.4, and the range from 1.15 to 1.17 times 4.166667 mm.sup.4 equals 4.79166705 mm.sup.4 to 4.87500039 mm.sup.4. The average moment of inertial of the cross sections 5-1-5-10 of the unit U1 is about 4.836496 mm.sup.4, which not only falls within the range from 4.75000038 mm.sup.4 to 4.91666706 mm.sup.4, but also falls within the range from 4.79166705 mm.sup.4 to 4.87500039 mm.sup.4.

[0034] In the second direction D2, one unit U2 of the first plate 21 is shown in FIG. 6. Assuming that the unit U2 shown in FIG. 6 is a square portion of 70.1 mm×70.1 mm, the thickness of the support portion 211 is 1 mm, the heights of the main portions 2121 and the branch portions 2122 of the embossing structures 212 formed on the support surface 2111 are 0.2 mm, and the unit U2 is cut into ten cross sections 6-1~6-10 along the second direction D2 with the same intervals, the moment of inertia of each of the cross sections 6-1~6-10 can be calculated based on formula of moment of inertia, $I = \int y^2 dA$, and is listed in Table 2.

TABLE-US-00002 TABLE 2 Cross sections Moments of inertia (mm.sup.4) 6-1 6.846542 6-2

6.731957 6-3 6.971736 6-4 6.971835 6-5 6.732086 6-6 6.846545 6-7 6.732086 6-8 6.971835 6-9

6.971736 6-10 6.731957 Average 6.850832

[0035] As for a flat plate having the same size as the unit U2 but without any embossing structure, the average moment of inertia of the cross sections of the flat plate in the second direction D2 is about 5.892557 mm.sup.4. As a result, in the second direction D2, the times of the average moment of inertia of the cross sections 6-1-6-10 of the unit U2 relative to the average moment of inertia of the cross sections of the flat plate having the same size as the unit U2 but without any embossing structure is about 1.163. In other words, in the case that the average moment of inertia of the cross sections of the flat plate in the second direction D2 is 5.892557 mm.sup.4, the range from 1.14 to 1.18 times 5.892557 mm.sup.4 equals the range from 6.71751498 mm.sup.4 to 6.95321726 mm.sup.4, and the range from 1.15 to 1.17 times 5.892557 mm.sup.4 equals 6.77644055 mm.sup.4 to 6.89429169 mm.sup.4. The average moment of inertial of the cross sections 6-1-6-10 of the unit U2 is about 6.850832 mm.sup.4, which not only falls within the range from 6.71751498 mm.sup.4 to 6.95321726 mm.sup.4, but also falls within the range from 6.77644055 mm.sup.4 to 6.89429169 mm.sup.4.

[0036] Compared the first plate 21 with the embossing structures 212 and the flat plate without any embossing structure which are suffered from a same force (e.g., 15 kgw), the times of the average moments of inertia of the cross sections of the first plate 21 with the embossing structures 212 in the first direction D1 and the second direction D2 are respectively 1.161 and 1.163, the downward bending deformation of the first plate 21 is improved about 10% compared to that of the flat plate without any embossing structure. In some specific load cases (e.g., non-uniform load cases), the

downward bending deformation of the first plate **21** may be improved up to 13%. Therefore, the first plate **21** with the embossing structures **212** can have a strong structural strength. In addition, even the overall thickness of the first plate **21** (e.g., the sum of the thickness of the support portion **211** and the thicknesses of the embossing structures **212**) and the thickness of the flat plate without any embossing structure are equal to each other (e.g., 1.2 mm), the first plate **21** with the embossing structures **212** can be manufactured with less material, such that the first plate **21** is lighter than the flat plate without any embossing structures, thereby meeting the lightweight requirement and carbon reduction requirement while satisfying the structural strength of the first plate **21**.

[0037] In the above paragraphs, the times of the average moments of inertia of the cross sections of the first plate **21** in the first direction **D1** and the second direction **D2** are obtained by comparing the flat first plate **21** with the flat plate without any embossing structure. In some other embodiments, when the first plate is a curved plate, the moment of inertia of the first plate may be obtained by projecting the first plate onto a plane and then performing calculation; that is, the curved plate may be transformed to the flat plate, and the embossing structures protruding from the support surface are the part to be considered only.

[0038] Note that the average moments of inertia of the cross sections of the units **U1** and **U2** of the first plate **21** in the first direction **D1** and the second direction **D2** are not restricted to being obtained from 10 cross sections. In some other embodiments, the average moments of inertia of the cross sections of the units **U1** and **U2** of the first plate in the first direction and the second direction may be obtained from another number of cross sections. For example, the average moments of inertia of the cross sections of the units **U1** and **U2** of the first plate in the first direction and the second direction may be obtained from 6 or 20 cross sections which are spaced apart from one another by the same intervals. In some examples, the intervals between adjacent two cross sections may not be greater than 2.5 mm. In a case that the average moments of inertia of the cross sections of the units **U1** and **U2** of the first plate in the first direction and the second direction are obtained from 6 cross sections, the average moments of inertia of the cross sections of the units **U1** and **U2** in the first direction and the second direction may be respectively 1.148 and 1.156. In a case that the average moments of inertia of the cross sections of the units **U1** and **U2** of the first plate in the first direction and the second direction are obtained from 20 cross sections, the average moments of inertia of the cross sections of the units **U1** and **U2** in the first direction and the second direction may be respectively 1.164 and 1.159. Therefore, it can be understood that the average moments of inertia of the cross sections of the units **U1** and **U2** of the first plate in the first direction and the second direction may be different due to the number of the cross-sections used in the calculation. Accordingly, during the design of the embossing structures of the first plate, the number of the cross sections may be selected to calculate the average moments in the first direction and the second direction according to the desired accuracy of the times of the average moments of inertia.

[0039] Note that the shapes of the embossing structures **212** of the first plate **21** are not restricted to the shapes shown in FIG. 3. As long as the times of the average moments of inertia of the cross sections of the first plate in the first direction and the second direction fall within the range from 1.14 to 1.18, the shapes of the embossing structures of the first plate may be modified randomly.

[0040] In addition, the times of the average moments of inertia of the cross sections of the first plate **21** are not restricted to falling within the range from 1.14 to 1.18 merely in the first direction **D1** and the second direction **D2**. In some other embodiments, when the first plate is also deformed obviously in a third direction (e.g., perpendicular to the first direction **D1**), the times of the average moments of inertia of the cross sections of the first plate in the first direction, the second direction and the third direction are required to fall within the range from 1.14 to 1.18.

[0041] Moreover, the times of the average moments of inertia of the cross sections of the first plate **21** in the first direction **D1** and the second direction **D2** are not restricted to both falling within the range from 1.14 to 1.18. In some other embodiments, when the design of the embossing structures

of the first plate mainly consider one direction, the time of the average moment of inertia of the cross sections of the first plate in that direction is merely required to fall within the range from 1.14 to 1.18.

[0042] In the aforementioned descriptions, the electronic device casing **20** is horizontally placed into the rack **10**, and two opposite sides of the electronic device casing **20** are fixed to the rack **10**, but the disclosure is not limited thereto. In some other embodiments, the electronic device casing may be vertically placed into the rack. In such a case, the times of the average moments of inertia of the cross sections of the first plate in other directions are required to fall within the range from 1.14 to 1.18.

[0043] Furthermore, in one embodiment, the times of the average moment of inertia of the cross sections of the first plate in one direction may be greater than 1.18.

[0044] Then, referring to FIG. **9**, FIG. **9** is a top view of an electronic device casing according to a second embodiment of the disclosure.

[0045] The electronic device casing **20a** of this embodiment is similar to the electronic device casing **20** of the previous embodiment, and the main difference between them is the shapes of the embossing structures of the first plate. Therefore, the following paragraphs merely introduce such difference, and the same parts between them will not be repeatedly introduced.

[0046] In this embodiment, each of embossing structures **212a** of a first plate **21a** of the electronic device casing **20a** includes a main portion **2121a** and four branch portions **2122a**. In each of the embossing structures **212a**, the main portion **2121a**, for example, has four recesses **21211a**, two of the recesses **21211a** are located at one side of the main portion **2121a** and are spaced apart from each other, and the other two of the recesses **21211a** are located at another side of the main portion **2121a** and are spaced apart from each other. Each of the branch portions **2122a** includes an end part **21221a** and a connection part **21222a** connected to each other. In each of the embossing structures **212a**, the two end parts **21221a** of two of the branch portions **2122a** respectively correspond to two of the recesses **21211a** located at one side of the main portion **2121a**, and the two end parts **21221a** of the other two of the branch portions **2122a** respectively correspond to two of the recesses **21211a** located at another side of the main portion **2121a**. As shown in FIG. **9**, the recesses **21211a** are symmetrically recessed inwards from two opposite long sides of the main portion **2121a** which substantially has a rectangular shape. In two of the embossing structures **212a** located adjacent to each other, the two end parts **21221a** of two of the branch portions **2122a** are connected to each other via the two connection parts **21222a**, and the two end parts **21221a** of other two of the branch portions **2122a** are connected to each other via the two connection parts **21222a**.

[0047] Referring to FIG. **10**, FIG. **10** is a partially enlarged view of a unit of a first plate along a first direction in FIG. **9**.

[0048] In the first direction **D1**, one unit **U1a** of the first plate **21a** is shown in FIG. **10**. Assuming that the unit **U1a** shown in FIG. **10** is a square portion of 50 mm×50 mm, the thickness of a support portion **211a** of the first plate **21a** is 1 mm, the heights of the main portions **2121a** and the branch portions **2122a** of the embossing structures **212a** formed on a support surface **2111** of the support portion **211a** are 0.2 mm, and the unit **U1a** is cut into ten cross sections 8-1~8-10 along the first direction **D1** with the same intervals, the moment of inertia of each of the cross sections 8-1~8-10 can be calculated by a computer software based on formula of moment of inertia, $I = \int y^2 dA$, and is listed in Table 3.

TABLE-US-00003 TABLE 3 Cross sections Moments of inertia (mm.^{sup.4}) 8-1 4.727699 8-2 4.798599 8-3 5.035815 8-4 5.036103 8-5 4.798598 8-6 4.727445 8-7 4.798598 8-8 5.036103 8-9 5.035815 8-10 4.798599 Average 4.879338

[0049] As for a flat plate having the same size as the unit **U1a** but without any embossing structure, the average moment of inertia of the cross sections of the flat plate in the first direction **D1** is about 4.166667 mm.^{sup.4}. As a result, in the first direction **D1**, the times of the average moment of

inertia of the cross sections 8-1-8-10 of the unit **U1a** relative to the average moment of inertia of the cross sections of the flat plate having the same size as the unit **U1a** but without any embossing structure is about 1.171.

[0050] Referring to FIG. 11, FIG. 11 is a partially enlarged view of a unit of the first plate along a second direction in FIG. 9.

[0051] In the second direction **D2**, one unit **U2a** of the first plate **21a** is shown in FIG. 11.

Assuming that the unit **U2a** shown in FIG. 11 is a square portion of 70.1 mm×70.1 mm, the thickness of the support portion **211a** is 1 mm, the heights of the main portions **2121a** and the branch portions **2122a** of the embossing structures **212a** formed on the support surface **2111a** are 0.2 mm, and the unit **U2a** is cut into ten cross sections 9-1~9-10 along the second direction **D2** with the same intervals, the moment of inertia of each of the cross sections 9-1~9-10 can be calculated based on formula of moment of inertia, $I = \int y^2 dA$, and is listed in Table 4.

TABLE-US-00004 TABLE 4 Cross sections Moments of inertia (mm.sup.4)

9-1	6.689948
9-2	6.708811
9-3	6.742207
9-4	6.742410
9-5	6.708788
9-6	6.689955
9-7	6.708788
9-8	6.742410
9-9	6.742207
9-10	6.708811
Average	6.718433

[0052] As for a flat plate having the same size as the unit **U2a** but without any embossing structure, the average moment of inertia of the cross sections of the flat plate in the second direction **D2** is about 5.892557 mm.sup.4. As a result, in the second direction **D2**, the times of the average moment of inertia of the cross sections 9-1-9-10 of the unit **U2a** relative to the average moment of inertia of the cross sections of the flat plate having the same size as the unit **U2a** but without any embossing structure is about 1.140.

[0053] Compared the first plate **21a** with the embossing structures **212a** and the flat plate without any embossing structure which are suffered from a same force (e.g., 15 kgw), the times of the average moments of inertia of the cross sections of the first plate **21a** with the embossing structures **212a** in the first direction **D1** and the second direction **D2** are respectively 1.171 and 1.140, the downward bending deformation of the first plate **21a** is improved about 8% compared to that of the flat plate without any embossing structure. Therefore, the first plate **21a** with the embossing structures **212a** can have a strong structural strength. In addition, even the overall thickness of the first plate **21a** and the thickness of the flat plate without any embossing structure are equal to each other (e.g., 1.2 mm), the first plate **21a** with the embossing structures **212a** can be manufactured with less material, such that the first plate **21a** is lighter than the flat plate without any embossing structures, thereby meeting the lightweight requirement and carbon reduction requirement while satisfying the structural strength of the first plate **21a**.

[0054] Referring to FIG. 12, FIG. 12 is a cross-sectional view of an immersion cooling system according to a third embodiment of the disclosure.

[0055] In this embodiment, the immersion cooling system **1000b** includes a container **10b** and an electronic device **5b**. The container **10b** is, for example, a tank and is configured to accommodate a coolant **C**. The electronic device **5b** is, for example, a server, and the electronic device **5b** is configured to be disposed in the container **10b** and at least partially immersed into the coolant **C**. The electronic device **5b** includes an electronic device casing **20b** and a motherboard **MB**.

[0056] In this embodiment, the electronic device casing **20b** is configured to be mounted in the container **10b**, where a first plate **21b** of the electronic device casing **20b** is, for example, vertically arranged in the container **10b**, and the first plate **21b** can be the same as the first plate **21** shown in FIG. 3 or the first plate **21a** shown in FIG. 9.

[0057] In this embodiment, the motherboard **MB** is located in an accommodation space **S** of the electronic device casing **20b**, and the first plate **21b** of the electronic device casing **20b** supports the motherboard **MB**. Specifically, the motherboard **MB** is hung and fixed on the first plate **21b**. The motherboard **MB** is provided with a heat generating module **CP**, comprising a heat source such as a chip (e.g., a CPU or a GPU) and a first power connector **PC1**. The electronic device **5b** may further include a power circuit board **PCB**, a cable **W** and a second power connector **PC2**. The power

circuit board PCB is, for example, disposed in the accommodation space S of the electronic device casing **20b**. The first power connector PC1 is connected to the power circuit board PCB via the cable W. The second power connector PC2 is, for example, disposed on the electronic device casing **20b** and located outside the electronic device casing **20b**. The power circuit board PCB is electrically connected to the second power connector PC2. The immersion cooling system **1000b** may further include a busbar **30b**, and the busbar **30b** is disposed in the container **10b**. The second power connector PC2 is assembled with the busbar **30b**.

[0058] In some embodiments, the immersion cooling system **1000b** is a two phase immersion cooling system and further comprises a condenser configured to be arranged above the coolant and condense vaporized coolant. In some embodiments, the condenser is configured to be arranged inside the container. In such embodiments, the heat generating module CP further comprises a boiling enhancing structure configured exposing to the coolant and absorb heat generated by the heat source. The boiling enhancing structure is a heat exchange structure used in an immersion cooling system. Its primary function is to rapidly transfer the heat generated by the heat source to a coolant. For example, in some immersion cooling systems, when the heat source is in operation, heat generated by the chip is conducted to the boiling enhancing structure, and the coolant vaporizes on a surface of the boiling enhancing structure and forms bubbles. Similar to bubbles generated when water boils, the bubbles continuously form and collapse, quickly dissipating the heat from both the boiling enhancing structure and the heat source. In some embodiments, the boiling enhancing structure may be a porous metal plate, and is directly attached to the heat source. In some embodiments, the heat generating module CP may further include a thermal interface, such as a thermal copper plate or a vapor chamber. The thermal interface can be disposed between the boiling enhancing structure and the heat source to provide excellent heat conduction and temperature uniformity.

[0059] According to the electronic device casing, the rack assembly and the immersion cooling system, the times of the average moment of inertia of cross sections of the first plate in one direction are not smaller than 1.14, which can meet the lightweight requirement and carbon reduction requirement while satisfying the structural strength of the first plate.

[0060] Furthermore, the times of the average moment of inertia of the cross sections of the first plate in one direction are not greater than 1.18, which can further increase the structural strength of the first plate so as to reduce the downward bending deformation amount of the first plate.

[0061] Preferably, the times of the average moment of inertia of the cross sections of the first plate in the first direction and the second direction fall within the range from 1.15 to 1.17, which can further increase the structural strength of the first plate so as to reduce the downward bending deformation amount of the first plate.

[0062] It will be apparent to those skilled in the art that various modifications and variations can be made to the present disclosure. It is intended that the specification and examples be considered as exemplary embodiments only, with a scope of the disclosure being indicated by the following claims and their equivalents.

Claims

1. A rack assembly, comprising: a rack; and an electronic device casing, horizontally or vertically mounted in the rack and comprising: a first plate, comprising at least one embossing structure, wherein times of an average moment of inertia of cross sections of the first plate in a first direction are not smaller than 1.14; and two second plates, connected to the first plate and forming an accommodation space with the first plate.
2. The rack assembly according to claim 1, wherein the times of the average moment of inertia of the cross sections of the first plate in the first direction are not greater than 1.18.
3. The rack assembly according to claim 1, wherein the at least one embossing structure comprises

a plurality of embossing structures, and the plurality of embossing structures are arranged in an array.

4. The rack assembly according to claim 2, wherein times of an average moment of inertia of cross sections of the first plate in a second direction are not smaller than 1.14 and are not greater than 1.18, and the second direction intersects the first direction.
5. The rack assembly according to claim 4, wherein the second direction is non-perpendicular to the first direction.
6. The rack assembly according to claim 2, wherein the times of the average moment of inertia of the cross sections of the first plate in the first direction are not smaller than 1.15 and are not greater than 1.17.
7. The rack assembly according to claim 1, wherein the two second plates are respectively mounted on two opposite sides of the rack, and the first direction is parallel to the two second plates.
8. An electronic device casing, comprising: a first plate, comprising: at least one embossing structure, wherein times of an average moment of inertia of cross sections of the first plate in a first direction are not smaller than 1.14; and two second plates, connected to the first plate and forming an accommodation space with the first plate.
9. The electronic device casing according to claim 8, wherein the times of the average moment of inertia of the cross sections of the first plate in the first direction are not greater than 1.18.
10. The electronic device casing according to claim 8, wherein the at least one embossing structure comprises a plurality of embossing structures, and the plurality of embossing structures are arranged in an array.
11. The electronic device casing according to claim 9, wherein times of an average moment of inertia of cross sections of the first plate in a second direction are not smaller than 1.14 and are not greater than 1.18, and the second direction intersects the first direction.
12. The electronic device casing according to claim 11, wherein the second direction is non-perpendicular to the first direction.
13. The electronic device casing according to claim 9, wherein the times of the average moment of inertia of the cross sections of the first plate in the first direction are not smaller than 1.15 and are not greater than 1.17.
14. The electronic device casing according to claim 8, wherein the at least one embossing structure has a main portion and a plurality of branch portions, the main portion has a plurality of recesses, the plurality of recesses are respectively located at two opposite sides of the main portion, and the plurality of branch portions correspond to the plurality of recesses.
15. The electronic device casing according to claim 14, wherein the plurality of recesses comprises four recesses, two of the four recesses are located at one side of the main portion and are spaced apart from each other, the other two of the four recesses are located at another side of the main portion and are spaced apart from each other, the plurality of branch portions comprises two branch portions, each of the two branch portions includes two end parts and a connection part connected to the two end parts, the two end parts of one of the two branch portions respectively correspond to two of the four recesses located at one side of the main portion, and the two end parts of the other one of the two branch portions respectively correspond to the other two of the four recesses located at another side of the main portion.
16. The electronic device casing according to claim 14, wherein the at least one embossing structure comprises a plurality of embossing structures arranged in an array; in each of the plurality of embossing structures, the plurality of recesses comprises four recesses, two of the four recesses are located at one side of the main portion and are spaced apart from each other, the other two of the four recesses are located at another side of the main portion and are spaced apart from each other, the plurality of branch portions comprises four branch portions, each of the four branch portions comprises an end part and a connection part connected to the end part, the two end parts of two of the four branch portions respectively correspond to two of the four recesses located at one

side of the main portion, the two end parts of the other two of the four branch portions respectively correspond to the other two of the four recesses located at another side of the main portion; in two of the plurality of embossing structures located adjacent to each other, the two end parts of two of the branch portions are connected to each other via the two connection parts, and the two end parts of other two of the branch portions are connected to each other via the two connection parts.

17. The electronic device casing according to claim 8, wherein the first direction is parallel to the two second plates.

18. An immersion cooling system, comprising: a container, configured to accommodate a coolant; and an electronic device, configured to be disposed in the container and at least partially immersed in the coolant and comprising: an electronic device casing, configured to be mounted in the container and comprising: a first plate, comprising at least one embossing structure, wherein times of an average moment of inertia of cross sections of the first plate in a first direction are not smaller than 1.14; and a motherboard, fixed on the first plate and provided with a heat generating module.

19. The immersion cooling system according to claim 18, further comprising a busbar disposed in the container, wherein the motherboard is further provided with a first power connector, the electronic device further comprises a power circuit board, a cable and a second power connector, the first power connector is electrically connected to the power circuit board via the cable, the second power connector is electrically connected to the power circuit board, and the second power connector is assembled with the busbar.

20. The immersion cooling system according to claim 19, wherein the power circuit board is disposed in an accommodation space of the electronic device casing, wherein the second power connector is disposed on the electronic device casing.

21. The immersion cooling system according to claim 18, wherein the times of the average moment of inertia of the cross sections of the first plate in the first direction are not greater than 1.18.

22. The immersion cooling system according to claim 18, wherein the first plate is vertically arranged in the container.

23. The immersion cooling system according to claim 18, wherein the heat generating module comprises a heat source and a boiling enhancing structure configured exposing to the coolant and absorb heat generated by the heat source.
